

# NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

First Semester B.Tech (CBCS) Degree Examinations

December 2023

CY1004-1 – CHEMISTRY OF CONSTRUCTION MATERIALS

(For CV)

Duration: 3 Hours

Max. Marks: 100

## Note:

**Part – A: Multiple Choice Questions:** Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

**Part – B: Descriptive Answer Questions:** Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

### PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- Nichrome is an example of \_\_\_\_\_ steel.
  - Heat-resisting
  - Magnetic
  - Tool
  - Spring
- The breaking, cracking, peeling off or fracturing of a refractory material when exposed to high temperature is known as \_\_\_\_\_.
  - Refractoriness
  - Thermal spalling
  - Thermal conductivity
  - Thermal expansion
- \_\_\_\_\_ is widely used in wire filaments for domestic lighting, industrial arc lamps and lighting.
  - Nickel
  - Molybdenum
  - Tungsten
  - Sodium
- A mixture of silica, lime and potash is the composition of \_\_\_\_\_ glass.
  - Boro-silicate
  - Soda-lime
  - Lead
  - Potash-lime
- The structure of \_\_\_\_\_ is semi-crystalline with netpoints.
  - SMA
  - SMP
  - Piezoelectric material
  - Electrochromic material
- Which of the following statement is NOT true?
  - SMA's take more time to recover to their original structure than SMPs.
  - SMPs are more employable in biomedical applications than SMA's
  - Piezoelectricity is reversible.
  - PZT is an example of piezoelectric material.
- Nanofilms is an example of \_\_\_\_\_ dimensional material.
  - Zero
  - One
  - Two
  - Three
- Chemical co-precipitation method is employed to synthesize \_\_\_\_\_ nanomaterials.
  - Graphene
  - Metal oxide
  - Pure metal
  - Metal hydroxide
- Pearl polymerization is also called as \_\_\_\_\_ polymerization.
  - Solution
  - Bulk
  - Suspension
  - Emulsion
- The monomers for the synthesis of nylon-6,6 are \_\_\_\_\_.
  - Adipic acid and hexamethylenediammine
  - Oxalic acid and tetraethylenediammine
  - Adipic acid and tetraethylenediammine
  - Oxalic acid and hexamethylenediammine
- Which of the below mentioned is NOT a property of carbon fibre?
  - High strength
  - Good corrosion resistance
  - Light weight
  - High co-efficient of thermal expansion

12. Epoxy resin is an example of \_\_\_\_\_ polymer.  
 A) Addition B) Condensation  
 C) Natural D) Crystalline
13. Choose the CORRECT statement.  
 A) Heavy rainfall improves soil fertility. B) Low temperature soils are more fertile.  
 C) Old soils are more fertile. D) Soil fertility is not influenced by the nature of parent rock.
14. Which of the following activity results in nitrogen loss in soil?  
 A) Nitrogen fixation B) Leaching  
 C) Manuring D) Ploughing
15. Complete the reaction:  $2\text{NO}_2 + \text{O}_2 \rightarrow$  \_\_\_\_\_  
 A)  $\text{N}_2\text{O}_2 + 2\text{O}_2$  B)  $2\text{NO}_3$   
 C)  $2\text{NO} + 2\text{O}_2$  D)  $\text{N}_2 + 3\text{O}_2$
16. Mottle leaf in plants is caused due to \_\_\_\_\_ deficiency.  
 A) Zn B) Fe  
 C) C D) Cl
17. Which of the following is NOT an example of reference electrode?  
 A) Glass electrode B) Standard hydrogen electrode  
 C) Calomel electrode D) Ag/AgCl electrode
18. What will be the potential developed in the cell:  $\text{Cu}/\text{Cu}^{2+}(5\text{M})//\text{Cu}^{2+}(5\text{M})/\text{Cu}$ ?  
 A) 5 V B) 25 V  
 C) 10 V D) 0 V
19. The potential of calomel electrode is dependent of the concentration of \_\_\_\_\_  
 A)  $\text{H}^+$  B)  $\text{Cl}^-$   
 C)  $\text{Na}^+$  D)  $\text{OH}^-$
20. Sensors with voltage as output, are classified as \_\_\_\_\_ sensors  
 A) Potentiometric B) Amperometric  
 C) Conductometric D) Colorimetric

### PART - B: DESCRIPTIVE ANSWER QUESTIONS

| Unit – I  |                                                                                                                                                                        | Marks | BT* | CO* | PO* |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|-----|-----|
| 1.        | a) What are the constituents of soda-lime glass? Elaborate the process of soda-lime glass manufacture,                                                                 | 9     | L*2 | 1   | 1   |
|           | b) Mention any four properties of ceramics. How are cement, concrete and mortar useful in construction?                                                                | 7     | L2  | 1   | 1   |
| 2.        | a) Compare and contrast shape memory alloys and shape memory polymers.                                                                                                 | 4     | L4  | 1   | 1   |
|           | b) Classify nanomaterials based on their dimension. Enlist any two engineering applications of carbon nanotubes.                                                       | 6     | L1  | 1   | 1   |
|           | c) Explain the sol-gel method of nanomaterial synthesis.                                                                                                               | 6     | L2  | 1   | 1   |
| 3.        | a) Mention the properties and applications of the following by giving a suitable example for each.<br>(i) Heat-resisting steel (ii) Spring steel (iii) Stainless steel | 9     | L2  | 1   | 1   |
|           | b) State the properties and engineering applications of graphene.                                                                                                      | 7     | L1  | 1   | 1   |
| Unit – II |                                                                                                                                                                        |       |     |     |     |
| 4.        | a) Write a note on suspension polymerization. Mention advantages and disadvantages of this method.                                                                     | 8     | L2  | 2   | 1   |
|           | b) Give the synthesis and applications of the following<br>(i) Carbon Fibre (ii) Epoxy resin                                                                           | 8     | L1  | 2   | 1   |
| 5.        | a) What is meant by soil productivity? Mention the factors affecting soil productivity.                                                                                | 8     | L2  | 2   | 1   |
|           | b) List out any four functions and deficiency symptoms of nitrogen in plants                                                                                           | 8     | L1  | 2   | 1   |

- |       |                                                                                                                                                                                               |   |    |   |   |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|---|---|
| 6. a) | What are syndiotactic polymers? Write the synthesis of chloro polyvinyl chloride.                                                                                                             | 3 | L1 | 2 | 1 |
| b)    | A sample of polymer contains 50% molecules with molecular mass 20,000, and 50% molecules with molecular mass 30,000. Calculate the number and weight average molecular weight of the polymer. | 6 | L3 | 2 | 2 |
| c)    | What are essential nutrients? Which are the elements present in Group I and Group II of essential nutrients and what are their biochemical functions?                                         | 7 | L1 | 2 | 1 |

**Unit – III**

- |       |                                                                                                                                                                                                                                                 |   |    |   |   |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|---|---|
| 7. a) | Explain the construction and working of glass electrode. How is it used to determine the pH of unknown solution?                                                                                                                                | 8 | L2 | 3 | 1 |
| b)    | Define sensor. Explain the theory underlying potentiometric sensors. How are they used in iron sensing?                                                                                                                                         | 8 | L2 | 3 | 1 |
| 8. a) | What is meant by concentration cell? A concentration cell was constructed by immersing two silver electrodes in 0.05 M and 0.1 M AgNO <sub>3</sub> solution. Write the cell reactions and calculate the EMF of the concentration cell at 298 K. | 7 | L3 | 3 | 2 |
| b)    | What is the basic principle governing conductometric sensors? Explain its working and state its applications.                                                                                                                                   | 9 | L2 | 3 | 1 |

BT\* Bloom's Taxonomy, L\* Level; CO\* Course Outcome; PO\* Program Outcome

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